

## Project Overview

### Project Details



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**Internship Program:** Cyberster Blue Team Internship

**Organization:** Cyberster

**Week 5 (Days 29–36)**

**Date:** July 20, 2026

**Status:** Completed

## Executive Overview

This project focused on analyzing a Windows malware sample using both static and dynamic malware analysis techniques. The malware sample, **Win32.AgentTesla**, was obtained from TheZoo malware repository and analyzed in a secure environment using multiple industry-standard tools. Static analysis was performed to examine the file without execution, while dynamic analysis was conducted using Hybrid Analysis to observe the malware's runtime behavior. The analysis confirmed that the sample was malicious and demonstrated behaviors including credential theft, persistence, process creation, and network communication.

## Sample Preparation

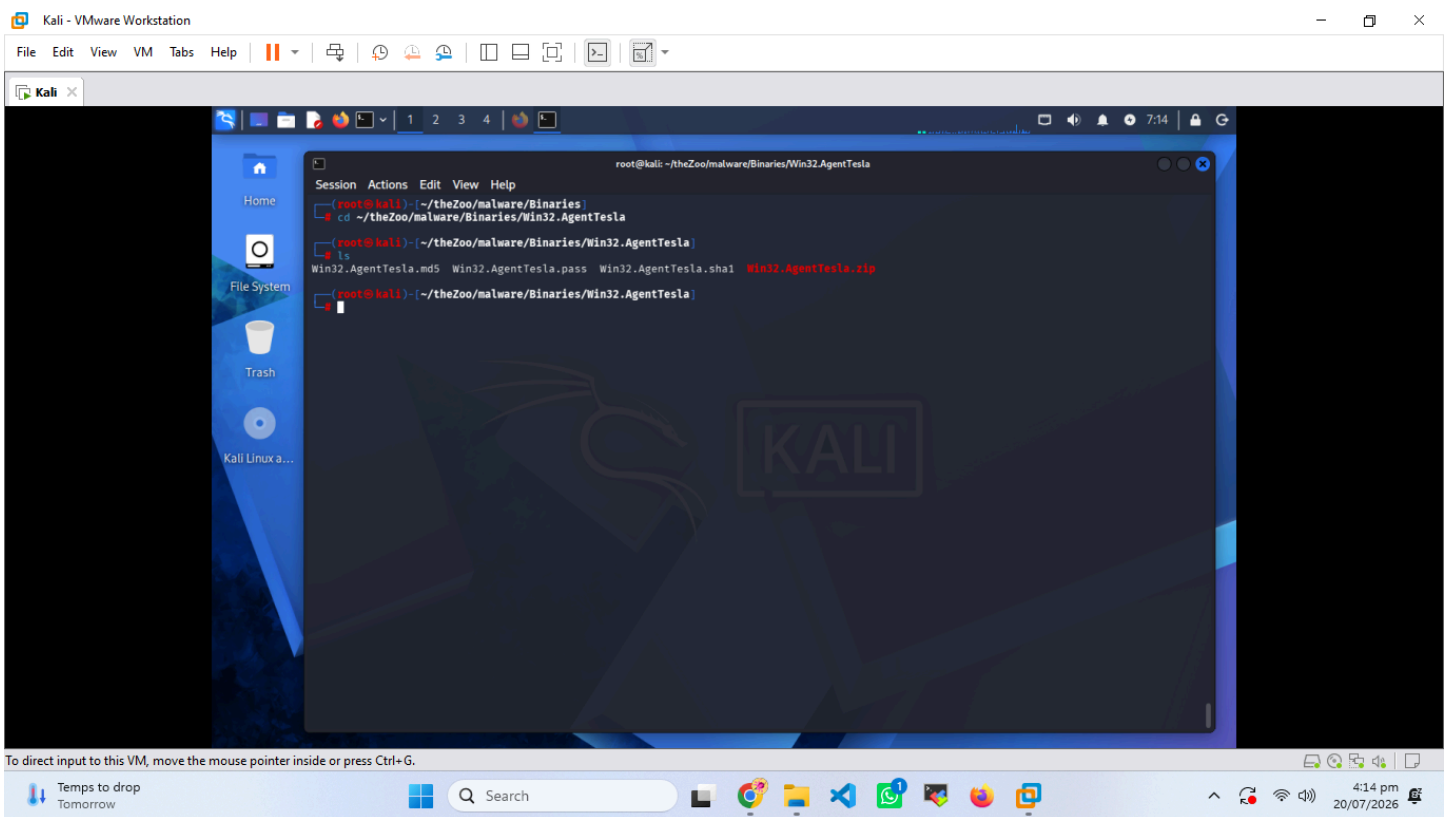
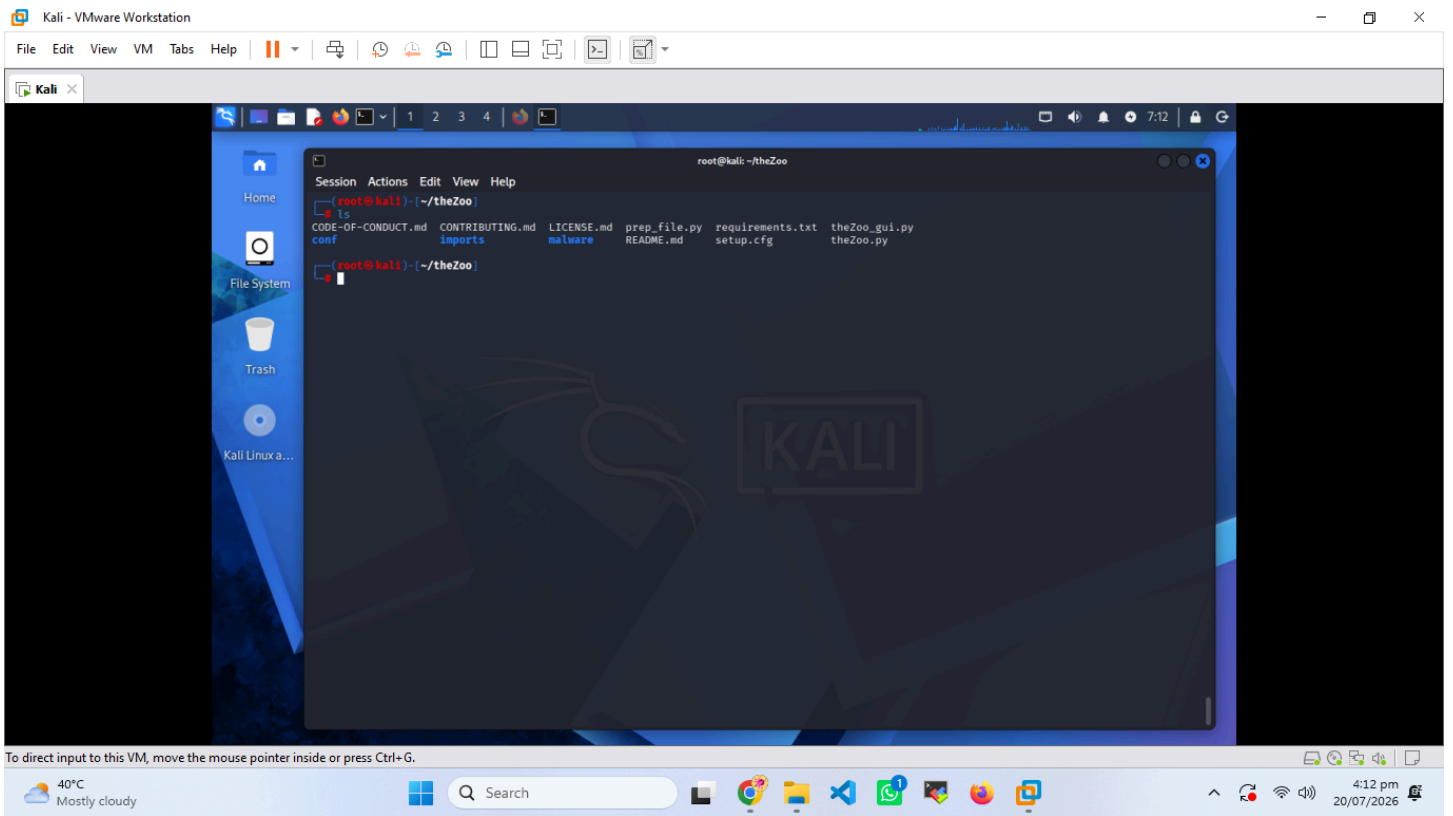
The first phase involved preparing the malware sample for analysis in a safe environment. TheZoo malware repository was cloned to Kali Linux, and the AgentTesla sample was selected. Since the malware was stored inside a password-protected ZIP archive, it was extracted safely using the provided password. After extraction, the integrity of the executable was verified by generating MD5 and SHA256 hashes.

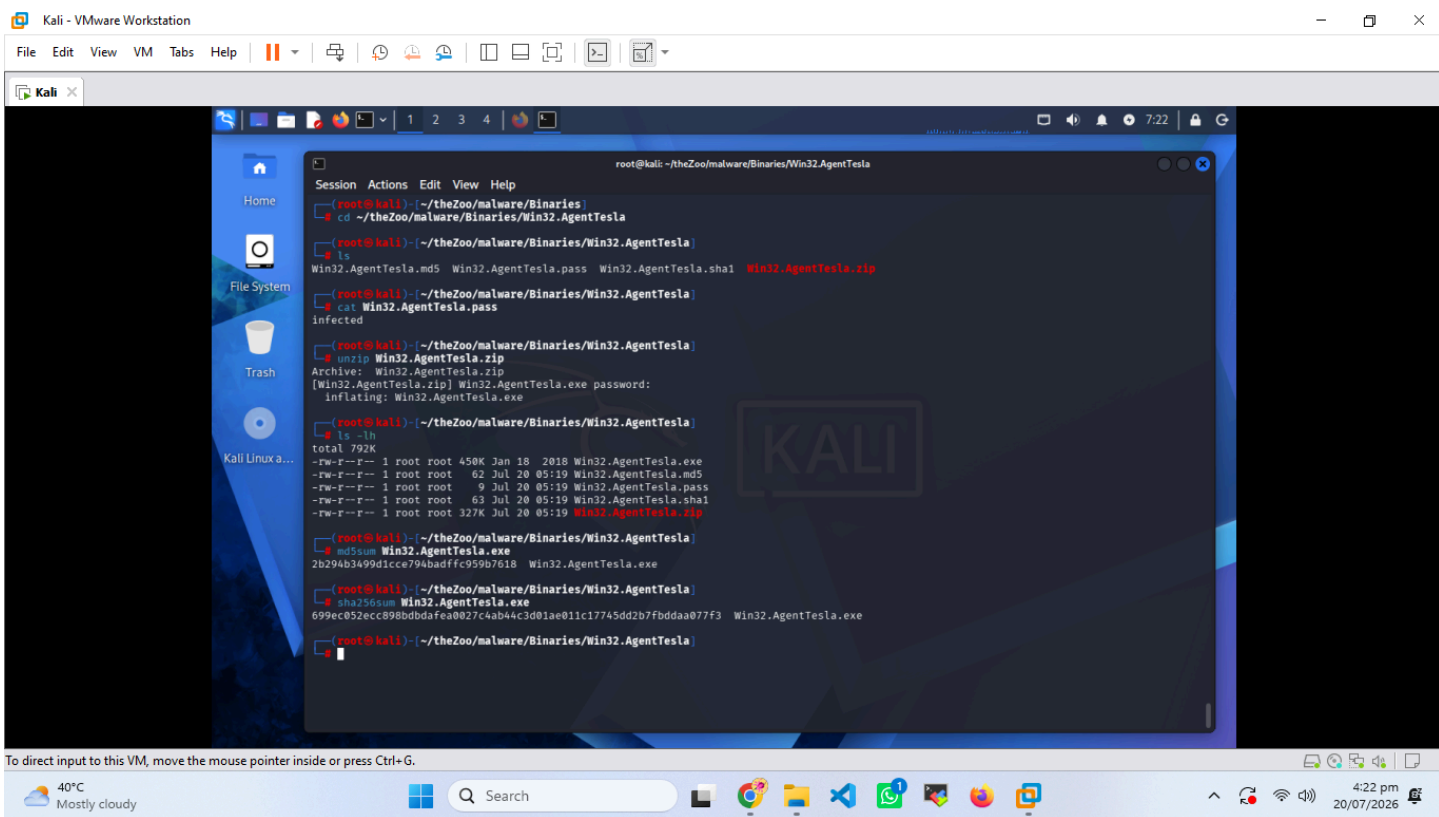
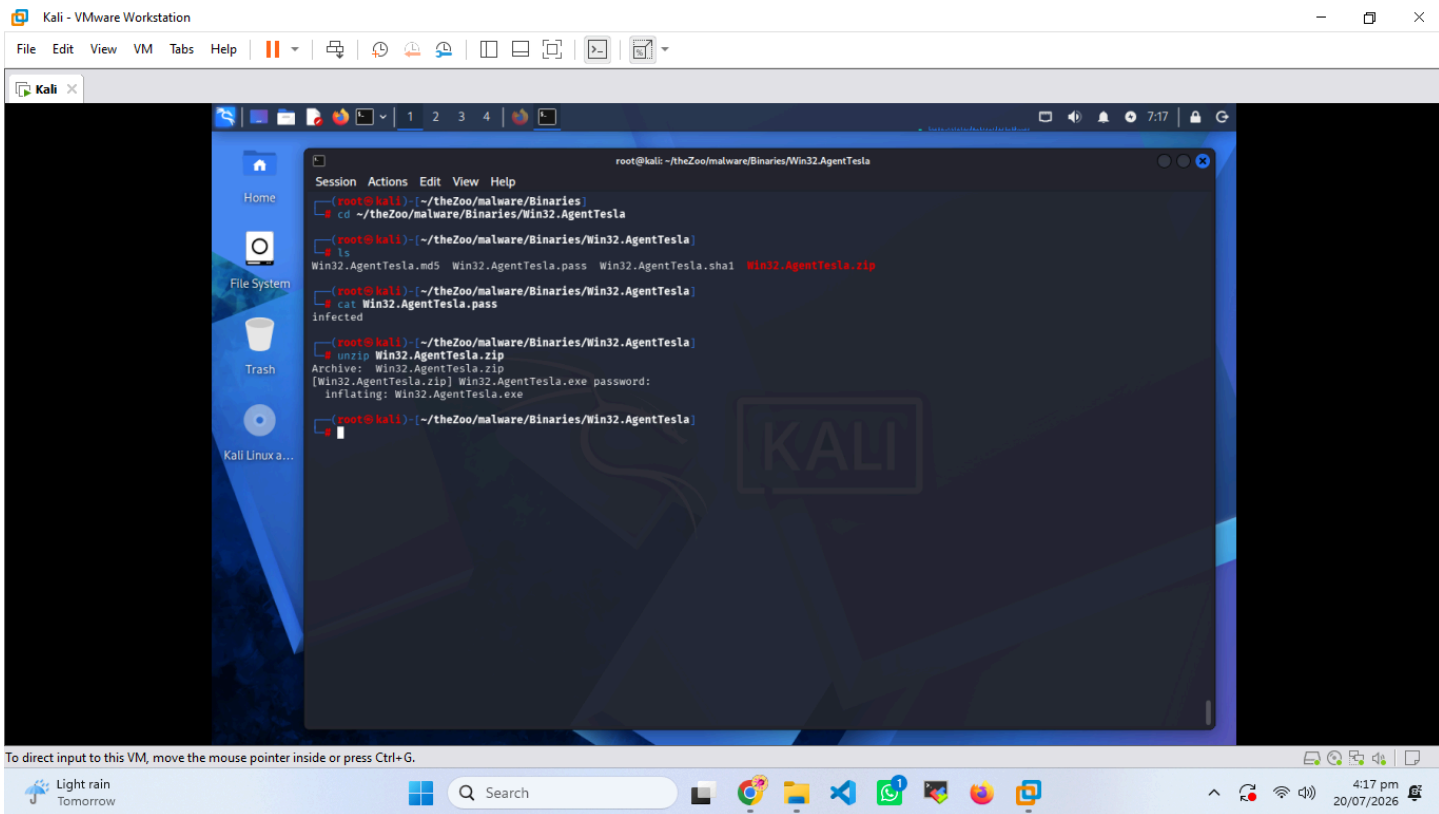
### Activities Performed

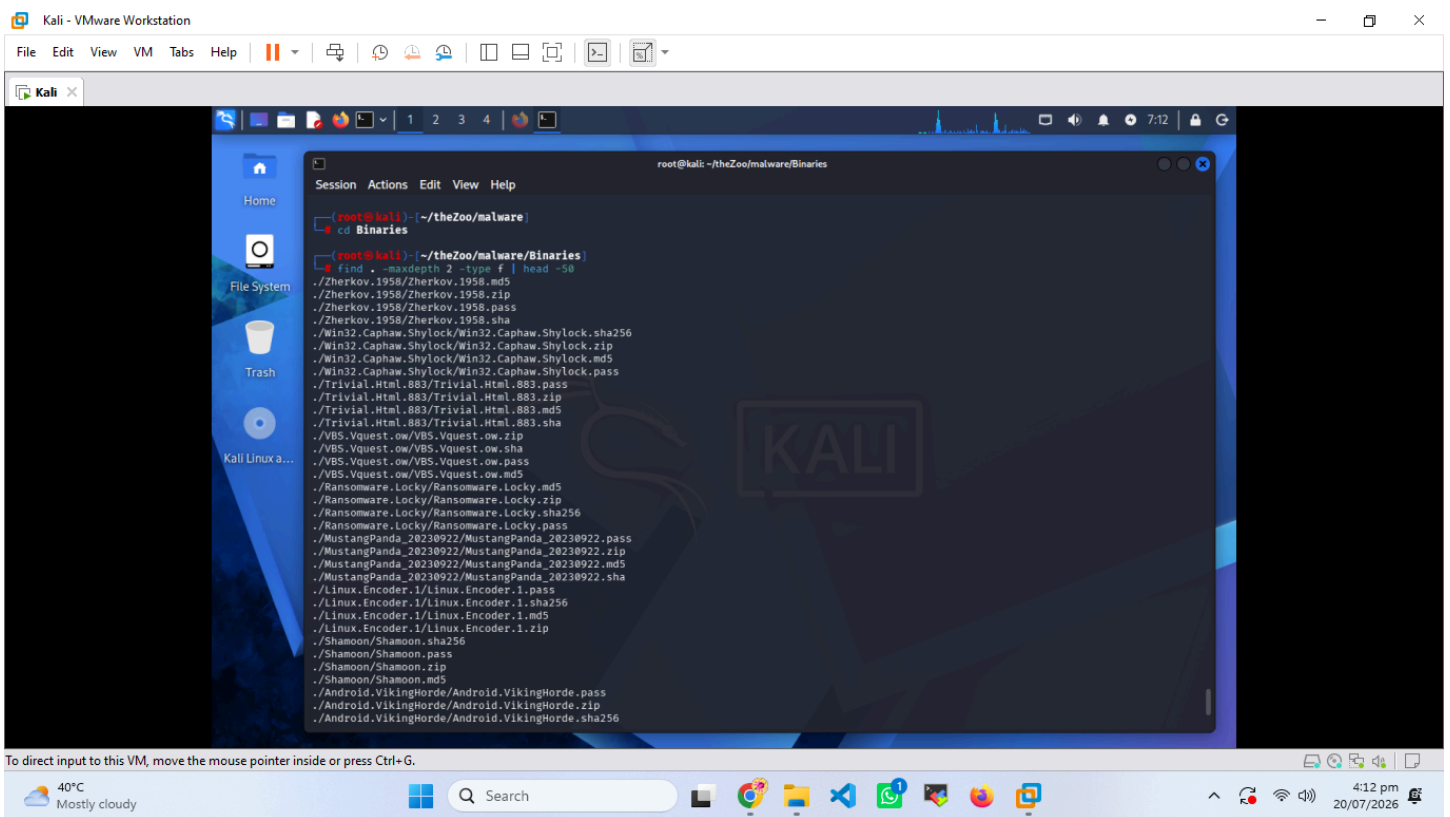
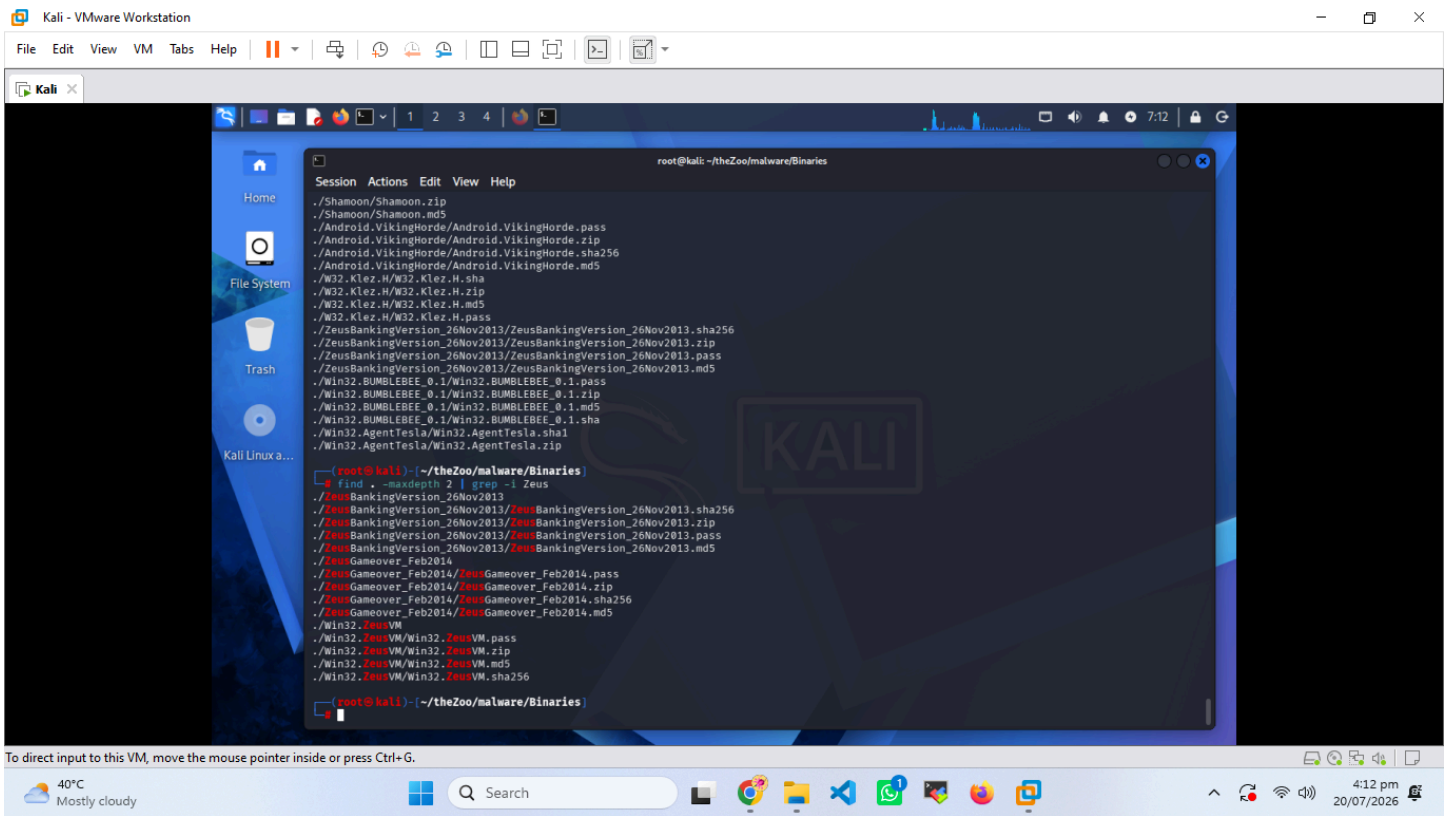
- Cloned TheZoo malware repository.
- Explored available malware samples.
- Selected **Win32.AgentTesla**.
- Extracted the password-protected ZIP archive.
- Generated MD5 and SHA256 hashes.

### Screenshots

1. **Downloaded TheZoo Repository**
2. **Browsing Malware/Binaries Directory**
3. **Selecting Win32.AgentTesla Sample**
4. **Extracting Password-Protected ZIP File**
5. **MD5 and SHA256 Hash Generation**







# Static Malware Analysis

Static malware analysis was performed without executing the sample. Multiple analysis tools were used to identify the executable type, metadata, embedded resources, readable strings, compiler information, and malware reputation.

## 2.1 ExifTool Analysis

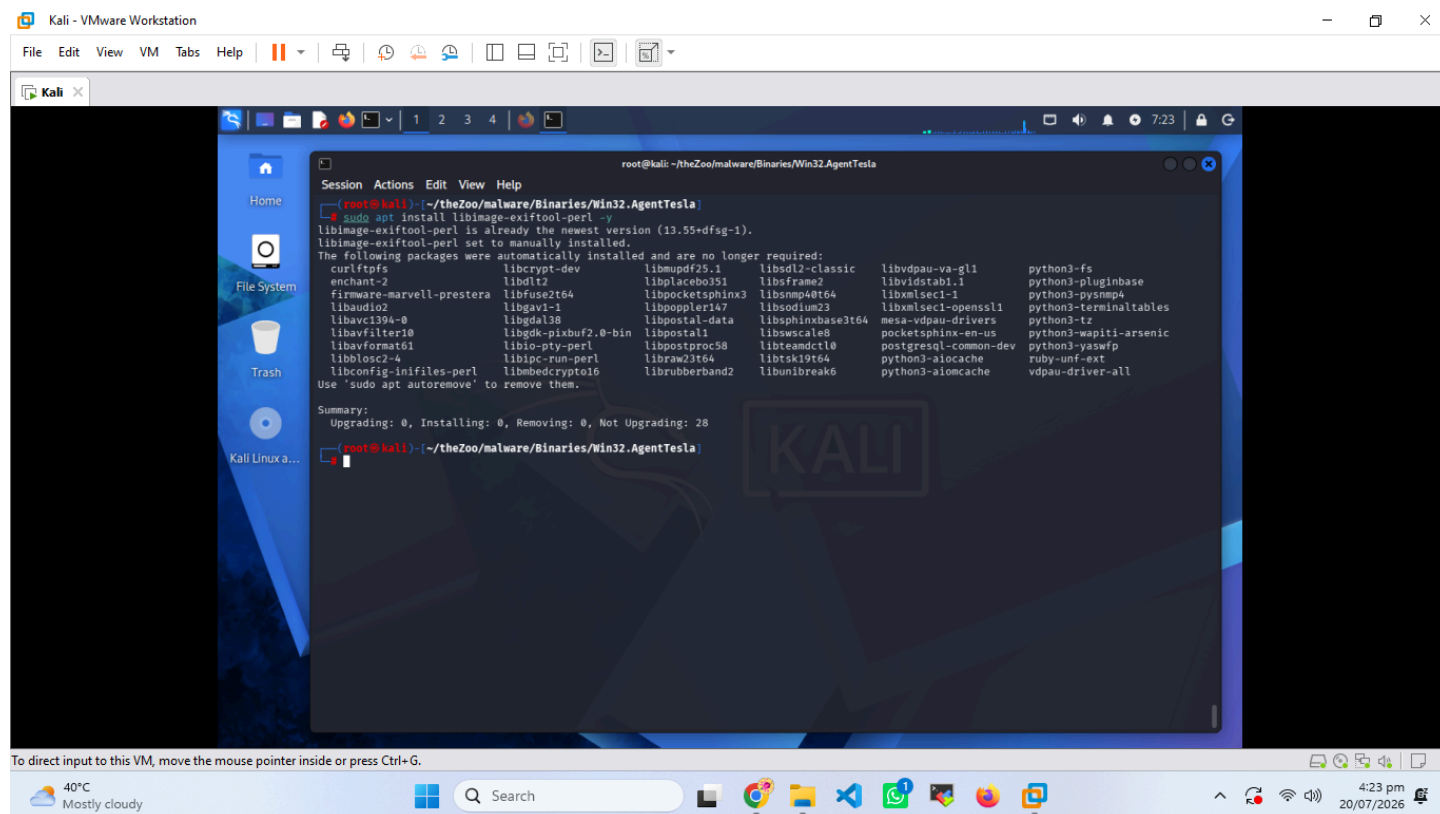
ExifTool was used to extract metadata from the executable.

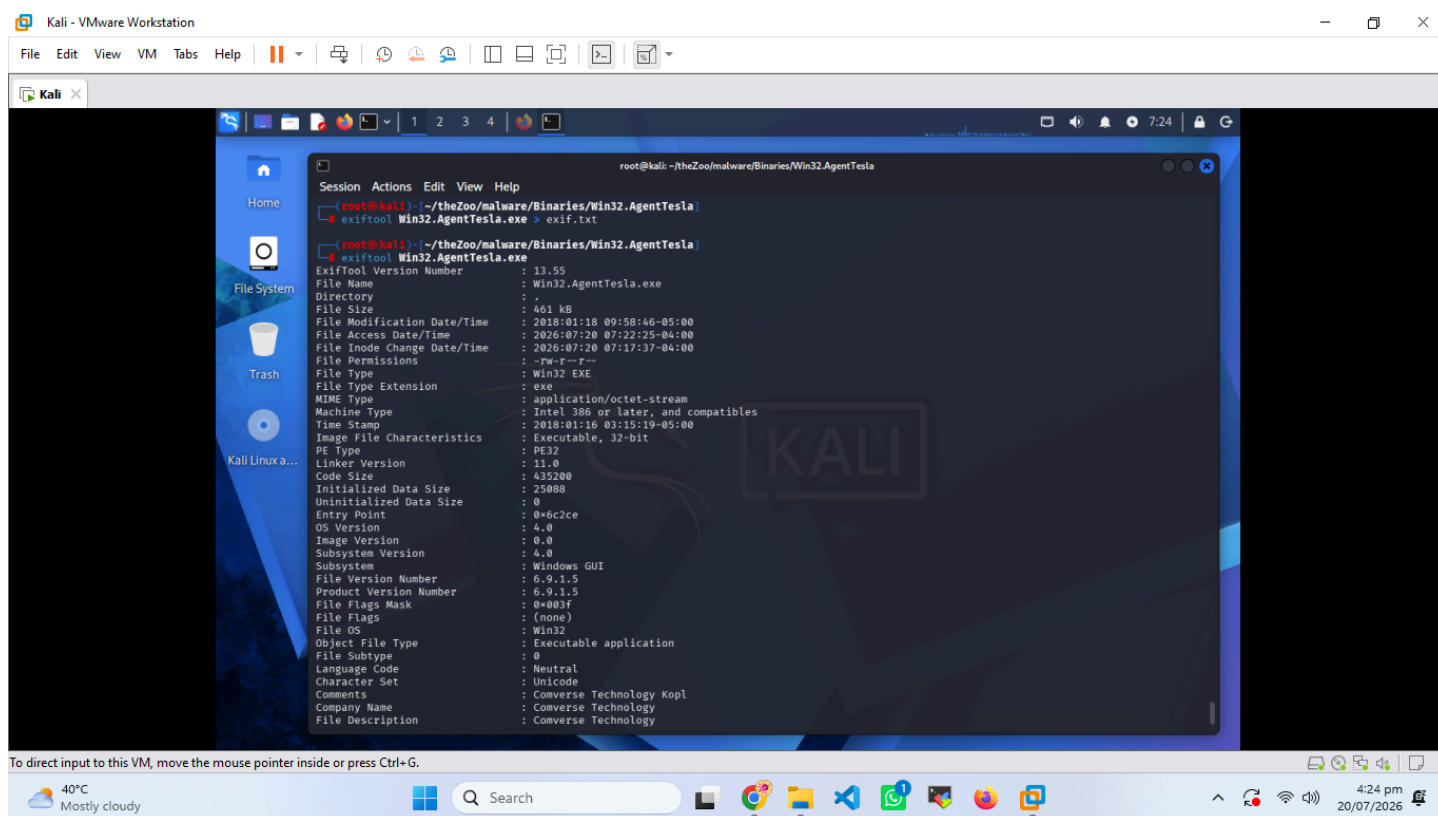
### Findings

- PE32 Windows executable.
- 32-bit application.
- Windows GUI subsystem.
- Original filename identified as **Po160118.exe**.
- Company information appeared to be spoofed.
- File version information was available.

### Screenshots

- Installing ExifTool
- ExifTool Metadata Output





## 2.2 Binwalk Analysis

Binwalk was used to inspect the executable for embedded resources.

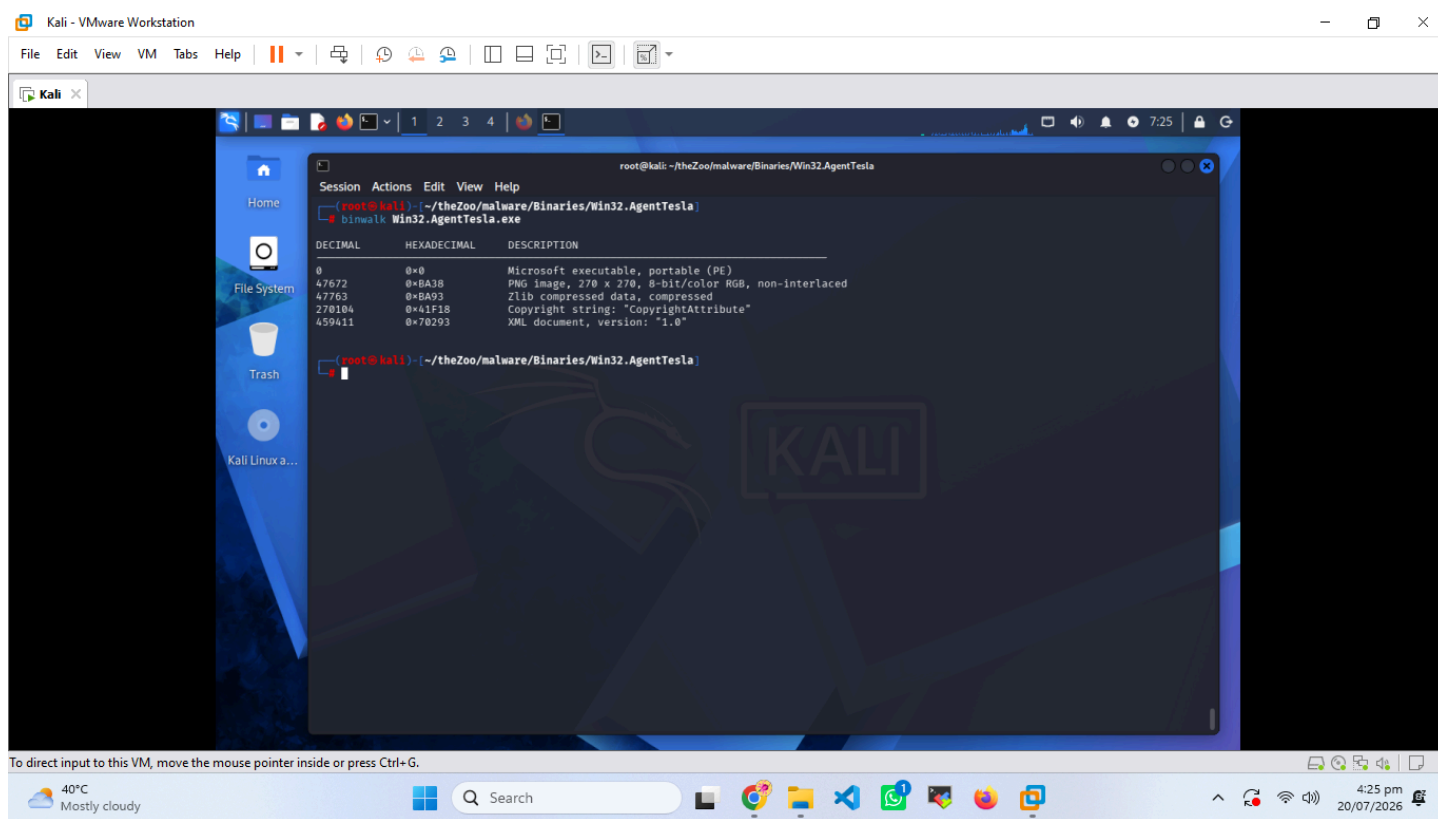
### Findings

- Embedded PNG resource detected.
- Zlib compressed data identified.
- XML resource found.
- No additional embedded executable payloads were observed.

### Screenshot

- Binwalk Output





## 2.3 Strings Analysis

The Strings utility extracted human-readable strings from the executable.

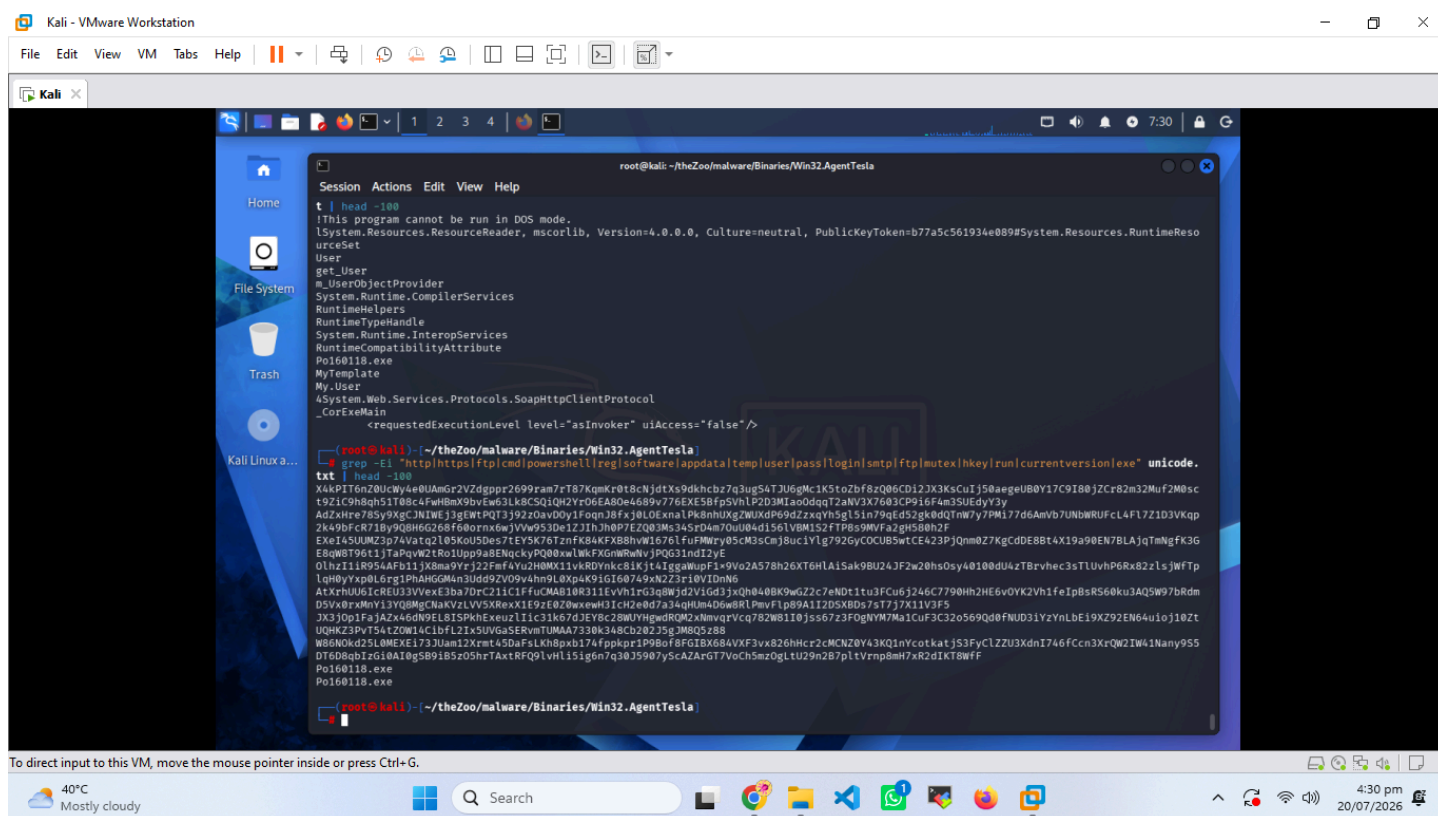
### Findings

- .NET Framework related strings identified.
- Internal filename Po160118.exe discovered.
- Limited readable indicators due to string obfuscation.

### Screenshot

- Strings Analysis Output





## 2.4 Detect It Easy (DIE)

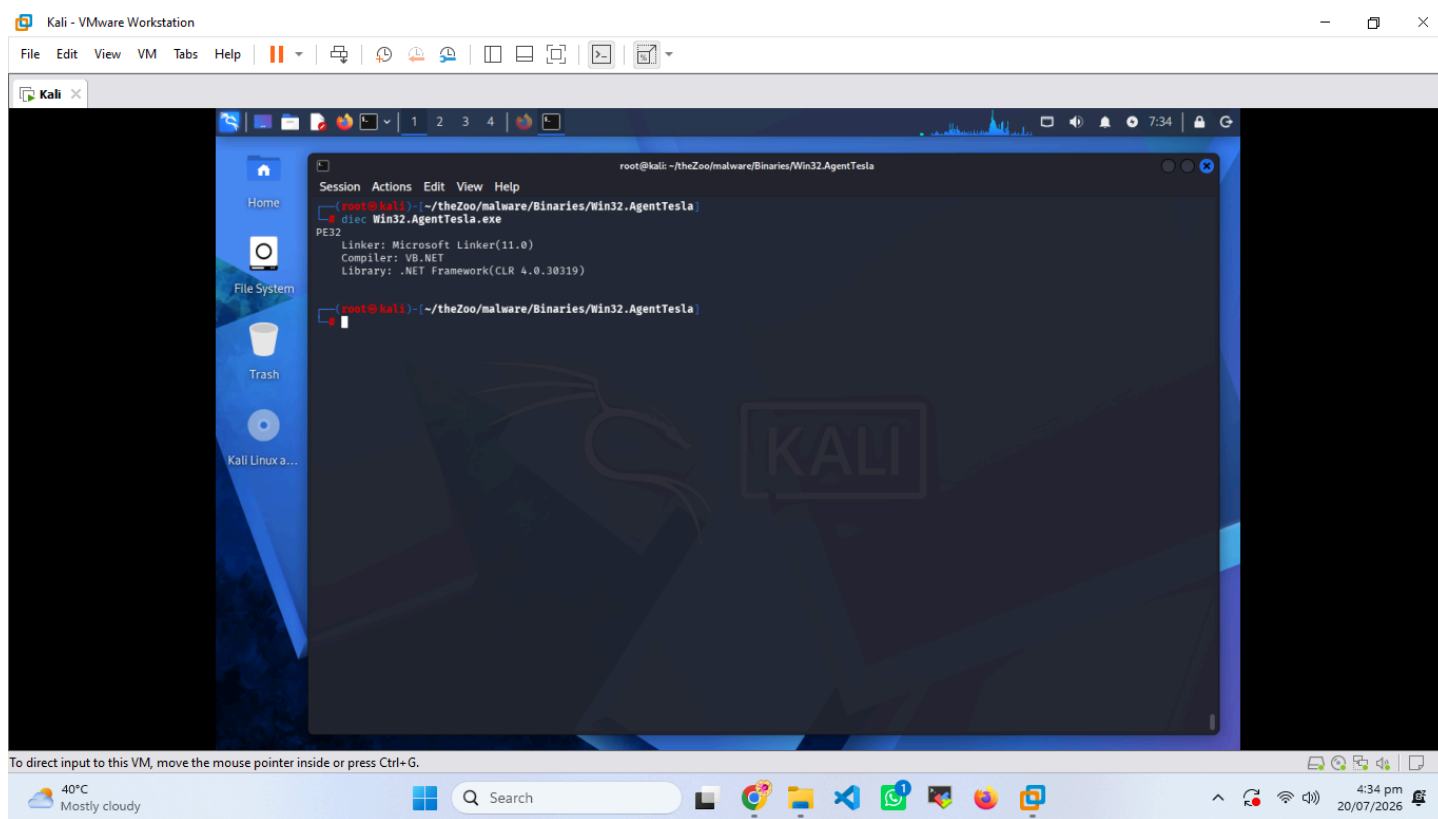
Detect It Easy was used to determine the executable type and compiler information.

### Findings

- PE32 Executable
- VB.NET Application
- .NET Framework (CLR 4.0)

### Screenshot

- Detect It Easy Result



## 2.5 VirusTotal Analysis

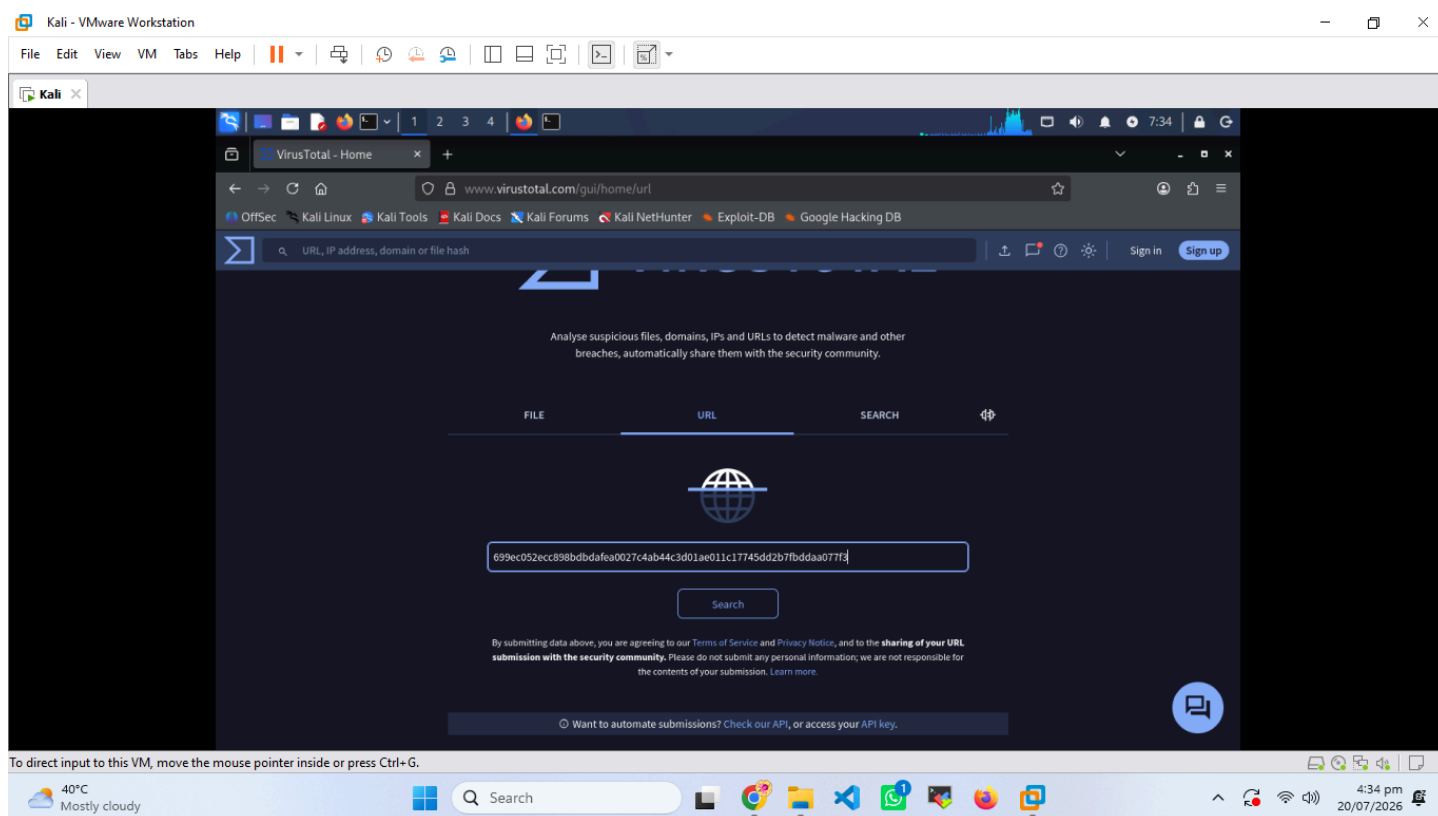
The SHA256 hash was submitted to VirusTotal to verify the reputation of the malware sample.

### Findings

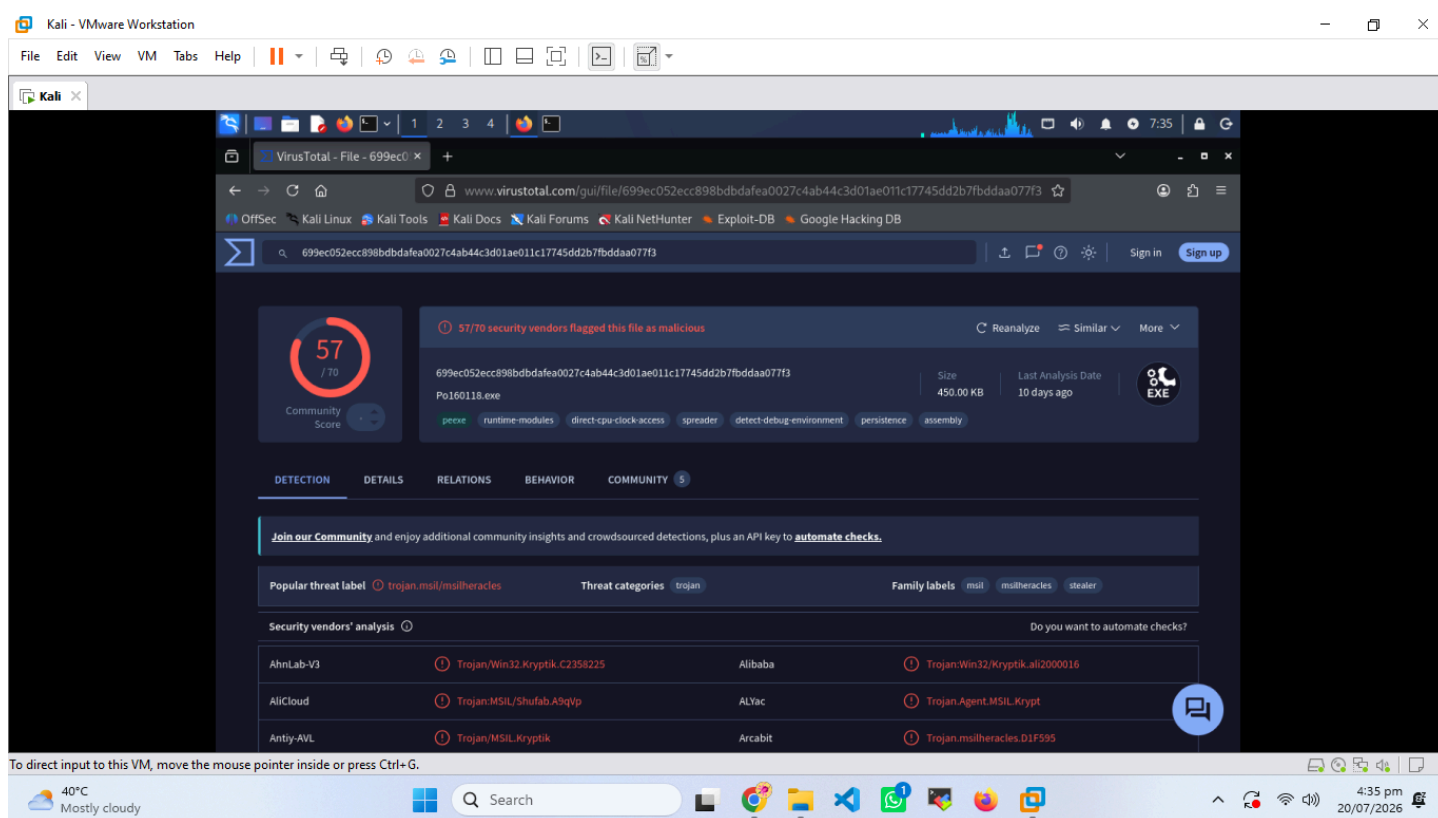
- 57 out of 70 security vendors detected the file as malicious.
- The sample was identified as a Trojan by multiple antivirus engines.

### Screenshots

- VirusTotal Hash Search
- VirusTotal Detection Result



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# Dynamic Malware Analysis

Dynamic analysis was performed using Hybrid Analysis to observe the malware's behavior during execution inside a secure sandbox environment.

## Hybrid Analysis Overview

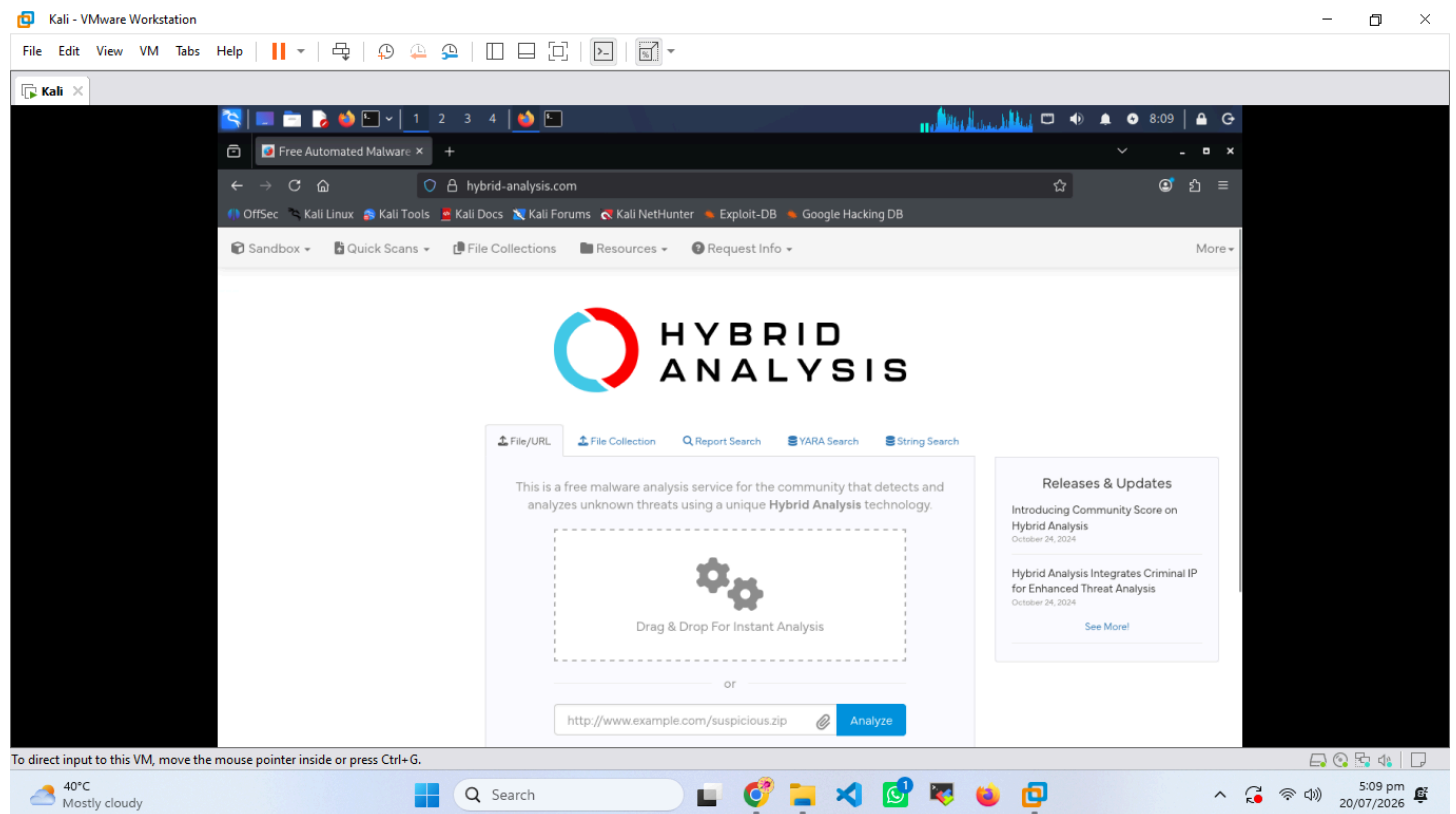
The sample was uploaded to Hybrid Analysis for behavioral analysis.

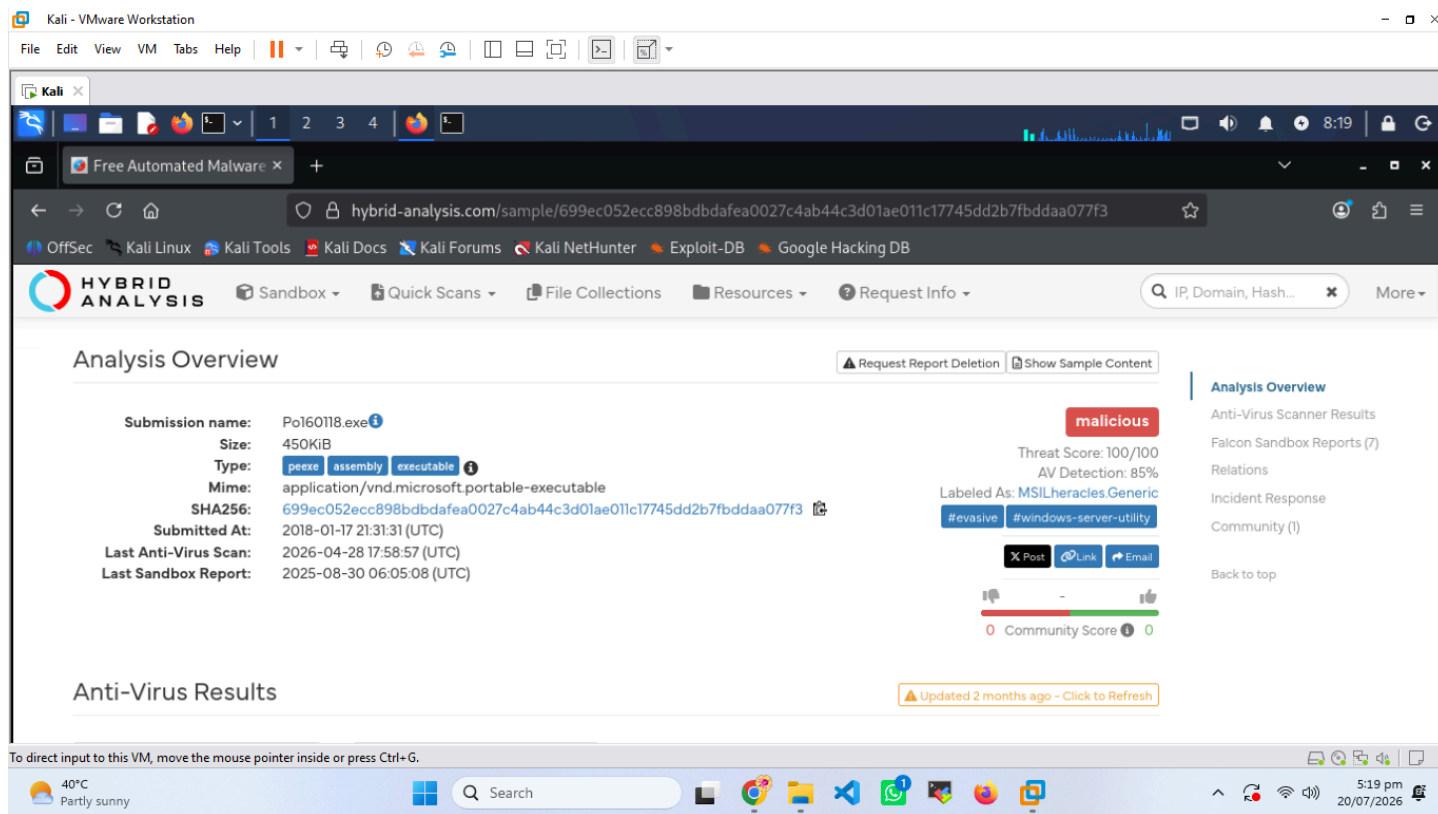
## Results

- Threat Score: **100/100**
- Antivirus Detection: **85%**
- Classification: **Malicious**

## Screenshots

- Hybrid Analysis Overview
- Threat Score and Antivirus Detection





## Malware Behavior Analysis

The sandbox revealed multiple malicious behaviors during execution.

### Observed Activities

- Attempted to access browser credentials.
- Read FTP client configuration files.
- Read FTP-related registry keys.
- Attempted to steal email credentials.
- Accessed PuTTY and WinSCP session information.
- Modified Windows Registry for persistence.
- Created additional processes.
- Performed process injection activity.
- Queried system information.
- Identified external IP address.
- Contacted external hosts.
- Dropped additional malicious files.

### Screenshots

- Incident Response Summary
- Behavior Analysis
- Relations / Dropped Files

Kali - VMware Workstation

File Edit View VM Tabs Help

Kali

Free Automated Malware x

hybrid-analysis.com/sample/699ec052ecc898bdbdfe0027c4ab44c3d01ae011c17745dd2b7fbdada077f3

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HYBRID ANALYSIS

Sandbox Quick Scans File Collections Resources Request Info

IP, Domain, Hash... More

### Falcon Sandbox Reports (7)

Characteristics Legend Show All As List Submit

Not all reports are visible. 2 error reports are hidden. Show All As List

|                                                                                                         |                                                                                                        |                                                                                        |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Windows 11 64 bit<br><b>Win32.AgentTesla.exe</b><br>August 30th 2025 06:05:08 (UTC)<br><b>Malicious</b> | Windows 10 64 bit<br><b>Win32_AgentTesla.exe</b><br>April 18th 2024 11:41:20 (UTC)<br><b>Malicious</b> | Windows 7 64 bit<br><b>xws.exe</b><br>July 4th 2018 06:28:23 (UTC)<br><b>Malicious</b> |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

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Kali - VMware Workstation

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Kali

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hybrid-analysis.com/sample/699ec052ecc898bdbdfe0027c4ab44c3d01ae011c17745dd2b7fbdada077f3

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HYBRID ANALYSIS

Sandbox Quick Scans File Collections Resources Request Info

IP, Domain, Hash... More

### Anti-Virus Results

Updated 2 months ago - Click to Refresh

|                                                                                                 |                                                                                 |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| CrowdStrike Falcon<br>Static Analysis and ML<br><b>Malicious (100%)</b><br>X No Additional Data | MetaDefender<br>Multi Scan Analysis<br><b>Malicious (18/26)</b><br>More Details |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|

CrowdStrike has donated Falcon MalQuery technology to power the "YARA search" and "String search" capabilities on Hybrid-Analysis. If you like what you see, you can easily upgrade to a full MalQuery license. It includes all features and access to the complete CrowdStrike malware repository consisting of over 1.2+

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Kali - VMware Workstation

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Kali x

Free Automated Malware x

hybrid-analysis.com/sample/699ec052ecc898bdbdfeaf0027c4ab44c3d01ae011c17745dd2b7fbdada077f3 80% ☆

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HYBRID ANALYSIS

Sandbox Quick Scans File Collections Resources Request Info

IP, Domain, Hash...

Incident Response

Risk Assessment

|                         |                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Remote Access</b>    | Tries to identify its external IP address                                                                                                                                                                                                                                                                                                                                   |
| <b>Spyware</b>          | <ul style="list-style-type: none"> <li>Accesses potentially sensitive information from local browsers</li> <li>Found a string that may be used as part of an injection method</li> <li>Tries to steal browser sensitive information (file access)</li> </ul>                                                                                                                |
| <b>Stealer/Phishing</b> | <ul style="list-style-type: none"> <li>Reads FTP client related files</li> <li>Reads FTP client related registry keys</li> <li>Scans for artifacts that may help identify the target</li> <li>Touched instant messenger related registry keys</li> <li>Tries to steal Mail credentials from registry</li> <li>Tries to steal Putty/Winscp sessions from registry</li> </ul> |
| <b>Persistence</b>      | <ul style="list-style-type: none"> <li>Creates new processes</li> <li>Modifies auto-execute functionality by setting/creating a value in the registry</li> <li>Spawns a lot of processes</li> <li>Writes data to a remote process</li> </ul>                                                                                                                                |

Analysis Overview  
Anti-Virus Scanner Results  
Falcon Sandbox Reports (7)  
**Relations**  
Incident Response  
Community (1)  
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Free Automated Malware x

hybrid-analysis.com/sample/699ec052ecc898bdbdfeaf0027c4ab44c3d01ae011c17745dd2b7fbdada077f3 80% ☆

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HYBRID ANALYSIS

Sandbox Quick Scans File Collections Resources Request Info

IP, Domain, Hash...

Incident Response

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fingerprint</b>      | <ul style="list-style-type: none"> <li>Spawns a lot of processes</li> <li>Writes data to a remote process</li> <li>Attempts to identify external IP address</li> <li>Queries kernel debugger information</li> <li>Queries process information</li> <li>Queries sensitive IE security settings</li> <li>Queries the display settings of system associated file extensions</li> <li>Reads the active computer name</li> <li>Reads the cryptographic machine GUID</li> <li>Reads the windows product ID</li> <li>Scans for artifacts that may help identify the target</li> <li>Tries to identify its external IP address</li> <li>Tries to steal Mail credentials from registry</li> </ul> |
| <b>Evasive</b>          | <ul style="list-style-type: none"> <li>Contains ability to terminate a process</li> <li>Executes WMI queries known to be used for VM detection</li> <li>Found a reference to a WMI query string known to be used for VM detection</li> <li>Marks file for deletion</li> <li>Possibly tries to evade analysis by sleeping many times</li> <li>Tries to hide tracks of having downloaded a file from the internet</li> <li>Tries to sleep for a long time (more than two minutes)</li> </ul>                                                                                                                                                                                               |
| <b>Ransomware</b>       | Shutdown/reboot a Windows system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Network Behavior</b> | Contacts 2 domains and 5 hosts. <a href="#">View all details</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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## Static Analysis Summary



| Tool           | Purpose                        |
|----------------|--------------------------------|
| ExifTool       | Metadata Analysis              |
| Binwalk        | Embedded Resource Detection    |
| Strings        | Readable String Extraction     |
| Detect It Easy | File Type & Compiler Detection |
| VirusTotal     | Malware Reputation Check       |

Dynamic Analysis Summary

| Observation               | Result   |
|---------------------------|----------|
| Threat Score              | 100/100  |
| Antivirus Detection       | 85%      |
| Browser Credential Access | Detected |
| FTP Credential Theft      | Detected |
| Registry Persistence      | Detected |
| Network Communication     | Detected |
| Dropped Files             | Detected |
| Virtual Machine Detection | Detected |

Key Findings

1. Successfully analyzed a PE32 Windows executable using static and dynamic techniques.
2. Metadata inspection identified the original filename and suspicious metadata.
3. Embedded resources and compressed data were detected.
4. VirusTotal confirmed the malware as malicious.
5. Hybrid Analysis assigned the maximum threat score.
6. The malware attempted to steal browser, FTP, email, and SSH credentials.
7. Registry modifications indicated persistence.
8. Network communication with external systems was observed.
9. Additional malicious files were created during execution.

## Recommendations

1. Never execute unknown executable files on a production system.
2. Perform malware analysis only inside isolated virtual machines or sandbox environments.
3. Verify file hashes before analysis.
4. Use multiple static and dynamic analysis tools to improve accuracy.
5. Monitor network traffic and registry changes during malware execution.
6. Keep antivirus signatures and threat intelligence feeds updated.

## Conclusion

This project successfully demonstrated the complete workflow of basic malware analysis using both static and dynamic techniques. Static analysis provided valuable information about the executable structure, metadata, compiler, and embedded resources without executing the malware. Dynamic analysis revealed the malware's runtime behavior, including credential theft, persistence, network communication, and system reconnaissance. Results obtained from VirusTotal and Hybrid Analysis confirmed that the sample was highly malicious. Overall, this project enhanced practical understanding of malware analysis methodologies and strengthened skills in identifying malicious behavior using industry-standard cybersecurity tools.

